נושאים מתקדמים בלמידת מכונה לביולוגיה חישובית

Advanced topics in Machine Learning for Computational Biology (0368-4238)

Homework 1: Epigenetic Methylation Clocks

Submission guidelines:

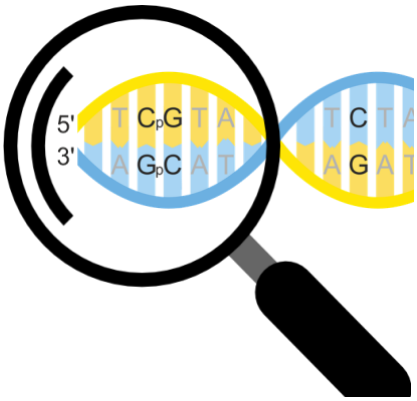
- The homework is due on **May 10th.**
- Homework is to be submitted alone or by pairs.
- Answers are to be submitted in English.
  Please submit your solution as a Jupyter or Google Colab notebook. Use markdown cells to write your answers to the questions and export the notebook as a pdf (with Google Colab: File -> Print -> Save as PDF).

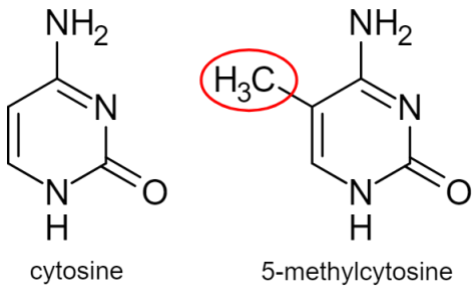
Guidelines on using LLMs (ChatGPT-like):

The purpose is (obviously) to learn about the tools studied in class rather than get a good grade ☺. As, such, LLM usage is only partially allowed:

- It is not permitted to directly prompt the model with the homework questions.
- Agents are not permitted (DeepSearch, DeepThink, Agent mode of Copilot, etc.).
- It is permitted to use LLMs to edit your answers for the sake of improving their readability.
- It is permitted to use LLMs for troubleshooting errors and code completion.
- It is permitted to use LLMs for generating working toy examples of the various functions called. However, you will learn more by spending time on the package documentations (scikit-learn, etc.).

Background:





DNA methylation is a biochemical process where a methyl group is added to the DNA molecule, typically at cytosine bases in CpG dinucleotides (see figure). It plays a key role in regulating gene expression and maintaining genomic stability. The **methylation level** of a CpG site is the proportion of DNA molecules in a sample that have a methyl group attached to the cytosine base at that site, often expressed as a **percentage (0 to 1)**. DNA methylation is detected using techniques like bisulfite sequencing or methylation arrays (e.g., Illumina Infinium) that identify methyl groups on cytosine bases, often after converting unmethylated cytosines to uracil.

DNA methylation and aging: In many sites, the methylation level of a CpG site is correlated (positively or negatively) with the chronological age. In general, the methylation level varies by tissue (brain, blood, etc.), and so does the correlation.

Epigenetic clocks: In 2013, Steve Horvath proposed the first "multi-tissue epigenetic methylation clock", a machine learning model that **predicts chronological age** from DNA methylation levels at specific CpG sites across the genome (measured at any tissue). The predicted age of an individual may be compared to its actual age to gain insights about his health. Link to the original article

Goal:

Our objective is to build an epigenetic methylation clock based on CpG methylation level measured from blood samples. This epigenetic age would be latter used as a biomarker in clinical settings. To make this test affordable, the model should only rely on a limited set of methylation sites (~50-100) while maintaining performance.

Dataset:

The dataset consists of measurements of DNA methylation levels from blood samples of ~11,910 donors with known age (between 0-109 years). Methylation levels were measured for ~ 450K sites using the Illumina 450K BeadChip platform – a technique cheap enough for research purpose but too expensive for routine clinical tests. The dataset was compiled from 19 separate studies by Varshavksy et al. Each site is identified by its name (e.g., "cg07559653"), which can be mapped to its location and sequence along the genome (e.g., Chromosome 16, base 73,589,978, reference gene: ZNRF1). **The 450K sites were already trimmed down to ~2375 CpG sites that are correlated with aging.**

A notebook is provided to load the dataset and conduct basic analysis.

Part I: Exploratory Data Analysis

For this section only, analyze the data using the entire dataset.

1. For each CpG site, calculate the Spearman correlation coefficient between its methylation level and the age of the donor. Display the result as a histogram and comment on the findings.

2. For the 5 CpG sites with highest and lowest Spearman correlation coefficient values, visualize the relationship between methylation level and age using a scatter plot. Comment on the findings.

3. Select all sites located along Chromosome 10 and reorder them by their location along the genome. Calculate the matrix of Spearman correlation coefficients between all pairs of methylation levels and visualize as a heatmap. Comment on the findings.

Part II: Splitting the data.

4. Report the number of samples per study and visualize the age distribution for each of the 19 studies, using a boxplot. Comment on the findings.

5. Read how the partition between train and test was done on the notebook.
    a. Why did we use dataset "GSE84727" as a test set rather than by using a random split?
    b. Why did we use a stratified split for the training set rather than a random split or a grouped split?

Part III: Training and evaluating a linear model

Throughout the rest of the homework, the metric for model selection and evaluation is chosen to be the Mean Absolute Error (MAE; $\ell(y, \hat{y}) = |y - \hat{y}|$).

Build a scikit-learn Pipeline consisting of:

- An imputer for missing values (SimpleImputer)
- Top-K feature selection using the F-statistics of the correlation coefficient (functions: SelectKBest and f_regression).
- Ridge Regression.

For each of the following values of K: [5,10,20,30,50,100,500,1000], select the optimal L2 regularization strength over the range by cross-validation.

6. Report the performance of the model on the test set as a table with error bars (defined as the standard deviation of the absolute error, divided by the square root of the test set size). How does the performance evolve with K?

We now focus on the model with K=50.

7. Plot the feature importance of the selected sites. Are some chromosomes over-represented among these sites?

8. For each of the five folds, the selected sites may be different. How many sites are found in all five folds? Discuss the findings.

Part IV: Automated feature selection by LASSO.

Replace the previous Pipeline by a Pipeline consisting of:

- An imputer for missing values (SimpleImputer)
- LASSO Regression.

9. For various values of the L1 regularization strength, fit a model (on the full training set) and calculate the performance on the test set as well as the number of **CpG sites with non-zero coefficients**. Plot the test set performance as function of the number of sites for both models.

   Why would feature selection with LASSO outperform feature selection by Pearson correlation? Conversely, why would Ridge outperform LASSO? Suggest a "best-of-both worlds" solution.

10. Repeat the analysis of Q8 with alpha=0.025. Are the findings different? Propose an explanation.

Part V: Experimenting with the loss function.

Since our target metric is the MAE rather than the mean square error, we might get better results by using it as a training loss function. This is implemented in two scikit-learn classes:

(1) QuantileRegression (with quantile=0.5) and L1 regularization. Exact, but slow optimization by linear programming.
(2) SGDRegression (with loss='huber',epsilon=1e-2) and L1/L2 regularization. Faster, approximate optimization by stochastic gradient descent. Pay attention that the learning rate needs to be adjusted.

11. Repeat question 6. with the MAE loss rather than Ridge regression and compare the findings.

Part VI: Training a Generalized Additive Model (GAM).

The exploratory data analysis showed a non-linear relationship between methylation level and age, suggesting that a GAM model may be more appropriate. To this end, build a Pipeline consisting of the following steps:

- An imputer for missing values (SimpleImputer)
- Top-K feature selection using the F-statistics of the correlation coefficient (functions: SelectKBest and f_regression).
- SplineTransformer using cubic splines, uniformly spaced knots, linear extrapolation.
- Ridge regression.

12. For each of the following values of K: [5,10,20,30,50], select the optimal regularization strength and number of knots, and report the performance over the test set. Compare the results with the ones of Part III.

13. Plot the feature effect functions using the Partial Dependency plot function of scikit-learn. Conclude on the benefits of using trainable non-linearities.

14. Do you expect that performance could be improved by instead implementing the GAM with boosted trees of depth 1?

Part VII: Glass-box vs Black-Box models.

15. Similarly train and evaluate a "black-box" Random Forest regressor on the same dataset. How does the performance compare with the one of the "glass-box" models? Conclude on the merits of each method.

Bonus: Suggest and implement another approach to further improve the performance of the age predictor.